Deep Learning for Medical Image Segmentation: From Convolutional Networks to Transformers — A Survey

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Abstract

Medical image segmentation is a cornerstone of modern clinical imaging, enabling precise delineation of anatomical structures and pathological regions from modalities such as computed tomography (CT), magnetic resonance imaging (MRI), and histopathology. Over the past decade, deep learning has fundamentally transformed this field, supplanting hand-crafted pipelines with end-to-end trainable architectures that approach or exceed human-level performance on challenging benchmarks. This survey provides a systematic review of deep learning methods for medical image segmentation, tracing the evolution from fully convolutional networks (FCNs) and the landmark U-Net architecture through attention-augmented convolutional models to the latest Vision Transformer (ViT) based frameworks. We categorise current approaches into three main families: (1) convolutional encoder-decoder networks, covering the U-Net family, DeepLab variants, and multi-scale feature pyramid methods; (2) hybrid CNN–Transformer architectures such as TransUNet and Swin-UNet that inject global context into local convolutional representations; and (3) pure 3D transformer models exemplified by nnFormer (Zhou et al., 2023), which interleaves convolution and self-attention with local and global volumetric attention. We also discuss weakly supervised and semi-supervised learning strategies that alleviate the high annotation cost characteristic of medical imaging. Key open challenges including limited labelled data, domain shift, 3D computational cost, and model explainability are identified, with promising future directions outlined. This survey is intended as a structured entry point for researchers entering the field and as a reference for practitioners designing segmentation systems.

Keywords: medical image segmentation, deep learning, convolutional

neural networks, transformer, U-Net, attention mechanism, semantic segmentation, volumetric segmentation

1. Introduction

Medical image segmentation—the process of partitioning an image into semantically meaningful regions—underpins a wide range of clinical workflows, from tumour volumetry and radiation treatment planning to surgical navigation and disease progression monitoring (Qureshi et al., 2023). Unlike natural scene segmentation, medical images present unique challenges: fine-grained boundary delineation is required at the voxel level; multiple imaging modalities (X-ray, CT, MRI, PET, ultrasound) possess heterogeneous noise characteristics; and obtaining pixel-level ground-truth annotations demands specialist knowledge that is scarce and expensive.

Classical segmentation pipelines relied on hand-crafted features combined with statistical or energy-minimisation models. Thresholding, active contours ("snakes"), graph cuts, and random forests were the dominant paradigms for two decades. While successful on constrained datasets, these methods generalise poorly across scanners, acquisition protocols, and pathology types.

The advent of deep learning changed the landscape dramatically. Long et al. (Long et al., 2015) introduced fully convolutional networks (FCNs), replacing the fully connected layers of classification networks with convolutional layers to produce dense pixel-wise predictions. Ronneberger et al. (Ronneberger et al., 2015) subsequently proposed U-Net, whose symmetric encoder-decoder architecture with skip connections became the *de facto* standard for biomedical image segmentation.

More recently, transformers—originally designed for natural language processing (Vaswani et al., 2017)—have entered computer vision (Dosovitskiy et al., 2021) and medical imaging. The self-attention mechanism's ability to model long-range dependencies across an entire image or volume addresses a fundamental limitation of convolution, which is inherently local. Hybrid models such as TransUNet (Chen et al., 2021) embed transformer blocks within U-Net-like encoders, while pure transformer approaches such as Swin-UNet (Cao et al., 2022) construct full encoder-decoder pipelines using shifted-window self-attention. For 3D volumetric data, nnFormer (Zhou et al., 2023) introduces interleaved convolution and self-attention with both local and global volume-based attention, significantly outperforming prior transformer baselines on organ segmentation benchmarks.

1.1. Paper Organisation

The remainder of this survey is organised as follows. Section 2 describes the research protocol. Section 3 covers background material including conventional methods and evaluation metrics. Section 4 reviews CNN-based approaches. Section 5 discusses attention mechanisms within convolutional frameworks. Section 6 covers transformer-based models. Section 7 addresses learning under limited annotation. Section 8 provides discussion including open challenges and future directions, and Section 9 concludes.

1.2. Research Highlights

The main contributions of this review are:

- A structured taxonomy of deep learning architectures for medical image segmentation covering three generations: FCN/U-Net, attention-augmented CNNs, and transformer-based models.
- A detailed treatment of nnFormer (Zhou et al., 2023), the state-of-the-art 3D transformer for volumetric segmentation, with architectural analysis and benchmark comparison.
- A comprehensive discussion of learning under limited annotation, including semi-supervised, weakly supervised, and self-supervised strategies specific to medical imaging.
- Identification of key open challenges (annotation cost, domain shift, 3D scalability, explainability) with concrete future research directions including foundation models and state-space models.

2. Research Protocol

2.1. Research Questions

This survey is guided by the following research questions:

RQ1. How have deep learning architectures for medical image segmentation evolved from FCN-based models to transformer-based models?

RQ2. What are the key architectural innovations that distinguish each generation of methods, and how do they address the limitations of prior approaches?

Table 1: Comparison with related survey papers.

Survey	**CNN**	**Attention**	**Transformer**	**3D**	**Semi-sup.**
Qureshi et al. (2023)	✓	✓	Partial	–	✓
This survey	✓	✓	✓	✓	✓

RQ3. Under what conditions do transformer-based models outperform CNN-based baselines, and what are the remaining gaps?

RQ4. What strategies exist for training effective segmentation models when labelled data is scarce?

RQ5. What are the most significant open challenges and promising research directions in this field?

2.2. Comparison with Existing Surveys

Several surveys have reviewed deep learning for medical image segmentation. Qureshi et al. (Qureshi et al., 2023) provide a comprehensive review of semantic segmentation methods for medical imaging, covering FCN-based, region- based, and weakly supervised methods with extensive application-specific discussion spanning skin cancer, breast cancer, retinography, brain tumour, and pulmonary nodules. Our survey complements this work with an in-depth focus on the transformer revolution post-2020, including systematic analysis of self-attention mechanisms, hybrid CNN–Transformer architectures, and the latest 3D volumetric transformer nnFormer. Table 1 summarises the coverage of key topics.

2.3. Analysis of Articles

The primary sources reviewed in this survey include papers published in leading venues including IEEE Transactions on Image Processing, IEEE Transactions on Medical Imaging, Medical Image Analysis, Information Fusion, MICCAI, CVPR, ICCV, and NeurIPS, covering the period 2015–2024. Search keywords used across PubMed, IEEE Xplore, and Google Scholar include: *medical image segmentation deep learning*, *U-Net variants*, *transformer medical imaging*, *3D segmentation volumetric*, and *semi- supervised medical segmentation*. The final reference list comprises 21 key papers selected for their architectural novelty, benchmark impact, and citation frequency.

3. Background

3.1. Conventional Segmentation Methods

Prior to deep learning, medical image segmentation relied on three broad families of algorithms.

Thresholding and region-based methods. Global or adaptive intensity thresholding partitions images based on pixel intensity alone. Seeded region growing (SRG) propagates labels from user-specified seed points to neighbouring pixels satisfying a homogeneity criterion (Qureshi et al., 2023). These methods are computationally efficient but sensitive to imaging noise and intensity inhomogeneity.

Active contour models. Snakes (Kass et al., 1988) represent boundaries as parametric curves minimising an energy combining internal smoothness constraints and external image-gradient forces. Level-set methods reformulate the problem implicitly, enabling topological changes such as contour splitting. These models provide smooth boundary estimates but require careful initialisation and hand-tuned parameters.

Graph-based methods. Graph cuts (Boykov and Jolly, 2001) formulate segmentation as a minimum-cut problem on a graph where edge weights encode pairwise boundary costs. These methods produce globally optimal solutions under their energy models but scale poorly to 3D volumes.

3.2. Imaging Modalities

Medical images exhibit modality-specific characteristics that influence algorithm design:

- **CT**: encodes Hounsfield units proportional to X-ray attenuation; bone appears bright, soft tissue grey, air dark.
- **MRI**: provides soft-tissue contrast via proton relaxation; different pulse sequences (T1, T2, FLAIR) highlight different tissues.
- **Histopathology**: H&E staining offers cellular-level resolution; colour variations encode tissue chemistry.
- **Ultrasound**: suffers from speckle noise and limited contrast; real-time acquisition enables interventional guidance.

3.3. Evaluation Metrics

Standard metrics for segmentation evaluation include:

Dice Similarity Coefficient (DSC):

$$\text{DSC} = \frac{2|P \cap G|}{|P| + |G|} \tag{1}$$

where P is the predicted mask and G the ground-truth mask. DSC ranges from 0 (no overlap) to 1 (perfect agreement).

Intersection over Union (IoU):

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|} \tag{2}$$

95th-percentile Hausdorff Distance (HD95): the 95th percentile of the directed Hausdorff distance between predicted and ground-truth surface points, measuring boundary accuracy.

Sensitivity and Specificity quantify true positive and true negative rates, critical in clinical applications where missed lesions carry different costs from false positives.

4. CNN-Based Architectures

4.1. Fully Convolutional Networks

Long et al. (Long et al., 2015) demonstrated that classification networks (AlexNet, VGG, GoogLeNet) could be converted to dense prediction networks by replacing fully connected layers with convolutional equivalents and using bilinear upsampling to restore spatial resolution. The key innovation was skip connections that fuse semantically rich deep-layer features with spatially precise shallow-layer features. FCN-8s, using three levels of skip connections, achieved state-of-the-art results on PASCAL VOC at the time and established the encoder-decoder paradigm underpinning virtually all subsequent work.

4.2. U-Net and Its Variants

U-Net (Ronneberger et al., 2015) was specifically designed for biomedical image segmentation where labelled data is scarce. Its symmetric encoder-decoder uses max-pooling for downsampling and transposed convolutions for upsampling, with *concatenation* skip connections (rather than addition)

Table 2: Summary of U-Net variants for medical image segmentation.

Model	Year	Key Contribution
U-Net (Ronneberger et al., 2015)	2015	Symmetric encoder-decoder with concatenation skip connections
3D U-Net (Çiçek et al., 2016)	2016	3D convolutions for volumetric segmentation from sparse annotation
U-Net++ (Zhou et al., 2018)	2018	Nested dense skip pathways; deep supervision
Attention U-Net (Oktay et al., 2018)	2018	Attention gates on skip connections to focus on relevant regions
nnU-Net (Isensee et al., 2021)	2021	Self-configuring pipeline; SOTA on 23 medical segmentation tasks

that preserve high-frequency spatial information in the decoder. Elastic deformation-based augmentation was proposed to compensate for limited training samples. Table 2 summarises major U-Net variants.

U-Net++ (Zhou et al., 2018) redesigns skip connections as nested, dense connections at the same resolution, allowing feature aggregation across multiple semantic scales. Deep supervision is applied at each decoder exit.

Attention U-Net (Oktay et al., 2018) places attention gates on skip connections. Given encoder feature $\mathbf{x}^l$ and decoder gating signal $\mathbf{g}$:

$$\alpha^l = \sigma_2\big(W_\psi \cdot \sigma_1(W_x\mathbf{x}^l + W_g\mathbf{g} + b_g) + b_\psi\big) \tag{3}$$

where σ_1 is ReLU, σ_2 is sigmoid, and $\alpha^l \in [0, 1]^{H\times W}$ is a spatial attention map applied to skip features before concatenation. This consistently improves segmentation of small structures such as the prostate and pancreas.

nnU-Net (Isensee et al., 2021) is a self-configuring framework that automatically adapts preprocessing, network topology, training, and post-processing based on dataset fingerprinting. Without manual architecture search, nnU-Net achieves state-of-the-art performance across a wide range of tasks and remains a strong competitive baseline against which new methods must be validated.

4.3. DeepLab and Multi-Scale Context

Chen et al. (Chen et al., 2018) introduced atrous (dilated) convolution, which enlarges the receptive field without increasing parameter count or sacrificing spatial resolution. DeepLab v3+ combines an encoder using Atrous Spatial Pyramid Pooling (ASPP)—parallel atrous convolutions at multiple rates—with a lightweight decoder that refines object boundaries. This multi-scale context aggregation is particularly effective for lesions spanning a wide range of sizes. Feature Pyramid Networks (FPN) (Lin et al., 2017) construct a top-down feature hierarchy with lateral connections, enabling simultaneous segmentation of structures at all scales.

5. Attention Mechanisms in CNNs

5.1. Channel and Spatial Attention

Squeeze-and-Excitation (SE) networks (Hu et al., 2018) compute per-channel attention weights by globally average-pooling spatial dimensions and passing the result through a bottleneck MLP, allowing the network to recalibrate feature responses. CBAM (Convolutional Block Attention Module) extends this with sequential channel-then-spatial attention, improving segmentation in densely labelled medical volumes.

5.2. Non-Local Networks

Non-Local Networks (Wang et al., 2018) compute pairwise affinities between all spatial positions, enabling global context modelling within a convolutional backbone. The non-local block is inserted as a plug-in module, incurring $\mathcal{O}(N^2)$ memory cost with spatial resolution N. Medical imaging applications include organ segmentation where global shape constraints matter.

5.3. Attention Gates in Skip Connections

As described in Section 4.2, Attention U-Net (Oktay et al., 2018) places spatial attention gates on skip connections. The attention weights are trained end-to-end via the segmentation loss without additional supervision, focusing the decoder on task-relevant spatial regions. Ablation studies confirm consistent 2–5% DSC improvements on challenging small-organ benchmarks.

6. Transformer-Based Architectures

6.1. Vision Transformer (ViT)

Dosovitskiy et al. (Dosovitskiy et al., 2021) demonstrated that a pure transformer applied to sequences of image patches achieves competitive classification accuracy when pre-trained on large datasets. An image of size $H \times W$ is split into $N = HW/P^2$ non-overlapping patches of size $P \times P$, each linearly projected to a D-dimensional embedding. A class token is prepended, learnable positional encodings are added, and the sequence is processed by L transformer encoder layers consisting of Multi-head Self-Attention (MSA) and MLP sublayers. ViT's global receptive field from the first layer contrasts with the local inductive bias of CNNs.

6.2. Swin Transformer

Liu et al. (Liu et al., 2021) introduced the Swin Transformer, addressing ViT's $\mathcal{O}(N^2)$ attention complexity by computing self-attention within non-overlapping local windows of size $M \times M$ and shifting the window partition between successive layers ("shifted window") to introduce cross-window connections. This reduces complexity to $\mathcal{O}(N)$ with respect to image tokens, making Swin practical for high-resolution medical images. Its hierarchical structure—patch merging reduces spatial resolution by $2\times$ across four stages—produces multi-scale feature maps compatible with dense prediction heads.

6.3. TransUNet: Hybrid CNN-Transformer

Chen et al. (Chen et al., 2021) proposed TransUNet, which uses a ResNet encoder to extract feature maps, then applies a ViT to the serialised feature map as a token sequence. The ViT-encoded global context is fed into a U-Net-like decoder with cascaded upsampling and skip connections. Compared to pure CNNs, TransUNet achieves improved DSC on multi-organ abdominal CT segmentation, demonstrating the benefit of combining local convolutional features with global transformer context.

6.4. Swin-UNet: Pure Transformer Encoder-Decoder

Cao et al. (Cao et al., 2022) built an encoder-decoder entirely from Swin Transformer blocks, introducing a symmetric Swin decoder with *patch expanding* operations that upsample feature maps by re-arranging tokens. Skip connections are retained analogously to U-Net. Swin-UNet achieves better

Table 3: Performance on the Synapse multi-organ CT segmentation benchmark (mean DSC %).

Method	**Mean DSC (%)**	**HD95 (mm)**
U-Net (Ronneberger et al., 2015)	74.68	36.87
TransUNet (Chen et al., 2021)	77.48	31.69
Swin-UNet (Cao et al., 2022)	79.13	21.55
nnFormer (Zhou et al., 2023)	**86.57**	**10.63**

performance than TransUNet on cardiac MRI and multi-organ CT segmentation, suggesting that hierarchical Swin features are more suitable than flat ViT features for dense prediction.

6.5. nnFormer: 3D Volumetric Transformer

Zhou et al. (Zhou et al., 2023) introduced nnFormer (not-another transFormer), specifically designed for 3D volumetric medical image segmentation. Three key innovations distinguish nnFormer from prior work.

Interleaved Convolution and Self-Attention. Rather than using convolutions only in the stem and transformers only at the bottleneck, nnFormer alternates convolutional blocks with transformer blocks throughout the encoder and decoder. This interleaving allows each layer to benefit from both local feature extraction (convolution) and global context (self-attention).

Volume-Based Local and Global Attention. nnFormer introduces two attention modes: (a) *local volume-based self-attention* that partitions the 3D feature map into non-overlapping cubic windows and computes self-attention within each window; (b) *global volume-based self-attention* using dilated windows covering the full volume, capturing inter-region dependencies. The two modes are applied alternately, analogous to the local-shifted mechanism in Swin Transformer but generalised to 3D.

Skip Attention. Traditional skip connections concatenate encoder and decoder features. nnFormer replaces concatenation with a skip attention block that applies cross-attention between encoder and decoder features, selectively retaining encoder details most useful for each decoder stage.

Table 3 summarises nnFormer's performance against representative baselines on the Synapse multi-organ CT benchmark.

nnFormer achieves an 86.57% mean DSC, outperforming Swin-UNet by more than 7 percentage points (Zhou et al., 2023). Its HD95 of 10.63 mm is less than half that of Swin-UNet (21.55 mm), indicating substantially more accurate boundary delineation.

7. Learning Under Limited Annotation

7.1. Semi-Supervised Learning

Semi-supervised methods leverage a small labelled set combined with a large unlabelled pool. Mean Teacher (Tarvainen and Valpola, 2017) maintains an exponential moving average (EMA) of model weights as a teacher; pseudo-labels from the teacher guide the student's training on unlabelled data. Uncertainty-guided filtering selects pseudo-labels with high teacher confidence, avoiding noisy supervision. Consistency regularisation enforces that strongly augmented views of the same image produce similar predictions, a principle particularly valuable in medical imaging where pixel-level annotations are expensive.

7.2. Weakly Supervised Learning

When only image-level labels or bounding-box annotations are available, Class Activation Maps (CAMs) from classification networks can generate rough segmentation seeds (Qureshi et al., 2023). Scribble-supervised methods accept sparse curve annotations from radiologists; conditional random field (CRF) post-processing then propagates labels to unlabelled pixels based on image gradients, achieving competitive results with a fraction of the annotation cost.

7.3. Self-Supervised Pre-Training

Contrastive learning (SimCLR, MoCo) and masked autoencoders (MAE) learn general visual representations from unlabelled data. Pre-trained models fine-tuned on small labelled medical datasets consistently outperform random initialisation, particularly for transformer models requiring large datasets to learn effective representations. Anatomical priors embedded in pre-training tasks—predicting relative positions of patches, reconstructing deliberately corrupted regions—further improve medical imaging performance.

8. Discussion

8.1. Current Limitations and Challenges

Limited and expensive annotation. Despite progress in semi- and weakly-supervised learning, most state-of-the-art models still require hundreds to thousands of annotated cases. The Segment Anything Model (SAM) (Kirillov et al., 2023), trained on over one billion masks from natural images, offers a promising zero-shot direction but requires domain adaptation to bridge the gap to medical images. Early adaptations (MedSAM, SAM-Med2D) show encouraging results.

Domain shift and multi-site generalisation. Intensity distributions vary across scanners, protocols, and imaging centres ("scanner bias"), causing models trained at one site to degrade at another (Qureshi et al., 2023). Federated learning enables collaborative training without sharing raw patient data, preserving privacy while improving generalisation.

Computational cost of 3D models. Extending attention to 3D volumes is memory-intensive. A 128^3 volume with patch size 4^3 produces 8,192 tokens; global self-attention requires $\mathcal{O}(8192^2)$ memory, prohibitive without special handling. Efficient variants—local windowed attention, linear attention, low-rank approximation—are active research areas. Hardware-aware design (e.g., FlashAttention) reduces memory footprint by avoiding materialisation of full attention matrices.

Explainability and clinical trust. Deep neural networks remain largely opaque. Gradient-based attribution (GradCAM, integrated gradients) provides coarse explanations; attention weights of transformer models offer a more natural interpretability pathway, though whether attention truly reflects reasoning is debated. Formal uncertainty quantification (deep ensembles, Monte Carlo dropout) is critical for flagging unreliable predictions in clinical deployment.

8.2. Future Directions

Foundation models for medical imaging. Large vision-language models accepting text, image, and clinical report inputs simultaneously are emerging. These models may unify segmentation, detection, and report generation within a single architecture.

Mamba and state-space models. The Mamba architecture (Gu and Dao, 2023), based on selective state-space models with linear complexity, has recently been applied to medical image segmentation (VMamba, SegMamba), offering an alternative to attention that scales better to high-resolution 3D inputs.

Synthetic data and generative augmentation. Diffusion models and GANs can synthesise realistic medical images with paired segmentation masks, alleviating data scarcity in rare pathologies and improving robustness to imaging variability.

Efficient architecture design. As clinical deployment moves to resource-constrained environments (edge devices, intraoperative systems), there is growing need for compact, fast segmentation models that maintain accuracy without requiring large GPU memory.

9. Conclusion

This survey has reviewed the evolution of deep learning methods for medical image segmentation, from the pioneering FCN and U-Net to state-of-the-art transformer- based architectures. Key milestones include: the encoder-decoder paradigm with skip connections (U-Net (Ronneberger et al., 2015)), the self-configuring pipeline (nnU-Net (Isensee et al., 2021)), hybrid CNN-Transformer designs (TransUNet (Chen et al., 2021)), pure Swin-Transformer encoder-decoders (Swin-UNet (Cao et al., 2022)), and the volumetric interleaved attention architecture (nnFormer (Zhou et al., 2023)).

The introduction of transformers has delivered consistent improvements on multi- organ and cardiac segmentation benchmarks. nnFormer achieves 86.57% mean DSC on the Synapse dataset—a 7 percentage-point improvement over Swin-UNet— confirming the value of combining local convolution and global self-attention in a unified 3D framework.

Challenges remain in annotation efficiency, domain generalisation, 3D computational scalability, and clinical explainability. Emerging paradigms including foundation models, self-supervised pre-training, state-space models, and diffusion-based augmentation offer compelling paths forward. As the field converges toward unified architectures that jointly process multimodal clinical data, deep learning for medical image segmentation is poised to become a standard component of clinical imaging workflows.

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